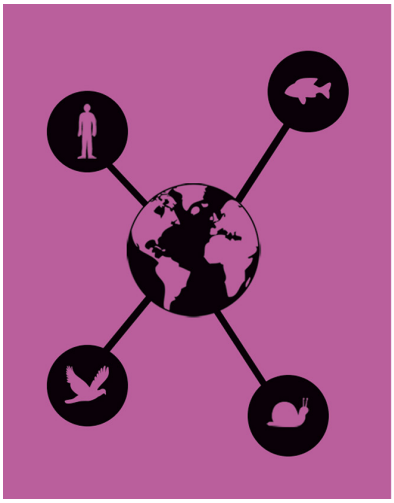


ALL LIFE IS CHEMICAL

ECOLOGICAL STOICHIOMETRY

IN CONTEXT

KEY FIGURES

Robert Sterner (1958–), **James Elser** (1959–)

BEFORE

1840 German biologist and chemist Justus von Liebig asserts that the limitations on agriculture productivity are primarily chemical.

1934 US oceanographer Alfred Redfield measures the atomic ratio of carbon, nitrogen, and phosphorus (C:N:P) in plankton and seawater, and finds it to be relatively consistent in all oceans. The Redfield Ratio soon becomes a benchmark for such research in all habitats.

AFTER

2015 In "Ocean stoichiometry, global carbon, and climate," Robert Sterner highlights inconsistencies in C:N:P ratios in phytoplankton, which absorb more atmospheric carbon in low-nutrient, low-latitude ocean surface waters and adjust their ratios accordingly.

Every living organism—from tiny ocean algae to a mighty redwood—is made up of chemical elements in varying ratios. Ecological stoichiometry considers the balance of these elements, and how the ratios change during chemical reactions. Studying such ratios throws light on the way the living